



***B.Tech. After 10th (3 years Diploma + 3 years Degree)
in Computer Science and Engineering -
Artificial Intelligence & Data Science***

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Polytechnic
Programme Name	B.Tech. After 10th (3 years Diploma + 3 years Degree) in Computer Science and Engineering - Artificial Intelligence & Data Science

+ + + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + + + + + + +

| Course Type | Description | % Credit Allotment | Credits Assigned |
|--|---|--------------------|------------------|
| Discipline Specific Core (DSC) | These courses are mandatory and provide the foundational knowledge within a student's chosen discipline at MIT-WPU | 41.27 | 52 |
| Discipline Specific Elective (DSE) | DSEs offer a pool of courses within a student's discipline, allowing them to specialize or explore related areas of interest. | 5.56 | 07 |
| Ability Enhancement Course (AEC) | AECs focus on enhancing language skills, communication, and other soft skills. | 23.81 | 30 |
| Skill Enhancement Course (SEC) | SECs provide hands-on training and practical skills to enhance employability and other practical applications of knowledge. | 7.94 | 10 |
| Value Education Course (VEC) | These courses aim to develop values, ethics, and societal awareness beyond the specific academic discipline. | 3.96 | 5 |
| Internship/ Apprenticeship/ Project/ Community (INP) | INP courses provide practical experience through internships, apprenticeships, research projects, or community engagement. | 14.29 | 18 |
| University Core (UC) | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. | 3.17 | 04 |
| Total | | 100 | 126 |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| I | AEC | Basic Mathematics | 4 |
| I | DSC | Basic Physics | 3 |
| I | SEC | Essentials of Computers | 2 |
| I | VEC | Yoga and Meditation | 1 |
| I | DSC | Graphics for Engineers | 2 |
| I | SEC | Web Page Designing | 3 |
| I | SEC | Computer Workshop Practices | 2 |
| I | DSC | Linux Basics | 3 |
| | | Total Credits: | 20 |
| I | UC | Social Leadership Development Program [SLDP] | 1 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| II | AEC | Applied Mathematics | 2 |
| II | AEC | Basic Electrical & Electronics Engineering | 4 |
| II | AEC | Programming in C | 4 |
| II | AEC | Communicative Competence | 3 |
| II | DSC | Data Communication and Computer Networks | 4 |
| II | VEC | Social and Life Skills | 1 |
| II | VEC | Environmental Education And Sustainability | 2 |
| | | Total Credits | 20 |
| II | UC | SPORTS | 1 |

First Year Exit

| Sr. No. | Course Type | Course name / Course Title | Credits |
|---------|-------------|----------------------------|----------|
| 1 | SEC | Web Page Designing | 3 |
| 2 | INP | Exit Internship 01 | 4 |
| | | Total Credits | 7 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| III | DSC | Database Management System | 3 |
| III | DSC | Digital Techniques And Microprocessor Programming | 4 |
| III | DSC | Data Structures using Python | 5 |
| III | DSC | Statistical Modelling For Machine Learning | 4 |
| III | DSC | Data Storytelling And Visualization | 3 |
| III | VEC | Essence Of Indian Constitution | 1 |
| | | Total Credits | 20 |
| III | AEC | Internet of Things | 1 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---------------------------------------|---------------|
| IV | SEC | UI/UX DESIGN | 3 |
| IV | AEC | Java Programming | 4 |
| IV | DSC | Operating System | 4 |
| IV | AEC | Mathematics For Machine Learning | 3 |
| IV | DSC | AI & ML Algorithms | 3 |
| IV | DSE | Natural Language | 3 |
| IV | DSE | Advanced Database Management | 3 |
| IV | DSE | Data Warehousing and Data Mining | 3 |
| | | Total Credits | 20 |
| IV | AEC | Discrete Mathematics and Graph Theory | 1 |
| IV | UC | CO-Creation | 1 |
| IV | UC | Foundation of Peace | 1 |

Second Year Exit

| Sr. No. | Course Type | Course Name / Course Title [Any one] | Credits |
|---------|-------------|--------------------------------------|----------|
| 1 | INP | Exit Industrial Training (Full Time) | 4 |
| 2 | SEC | Second Year Exit Course(Online) | 4 |
| | | Total Credits | 4 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| V | INP | Capstone Project | 2 |
| V | DSC | Software Engineering | 4 |
| V | DSC | Cloud Computing for Data Science | 3 |
| V | DSC | Mobile Application Development | 4 |
| V | DSC | Big Data Analytics | 3 |
| V | DSE | 1) Advanced Machine Learning
2) Image Processing
3) Business Intelligence | 4 |
| | | Total Credits: | 20 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| VI | AEC | Entrepreneurship Development & Startups | 2 |
| VI | AEC | Management | 2 |
| VI | INP | Internship | 15 |
| VI | INP | Project Seminar | 1 |
| | | Total Credits: | 20 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.